

PRE-COVID (2019) **SOFIA REAL ESTATE MARKET CONDITIONS ANALYSIS**

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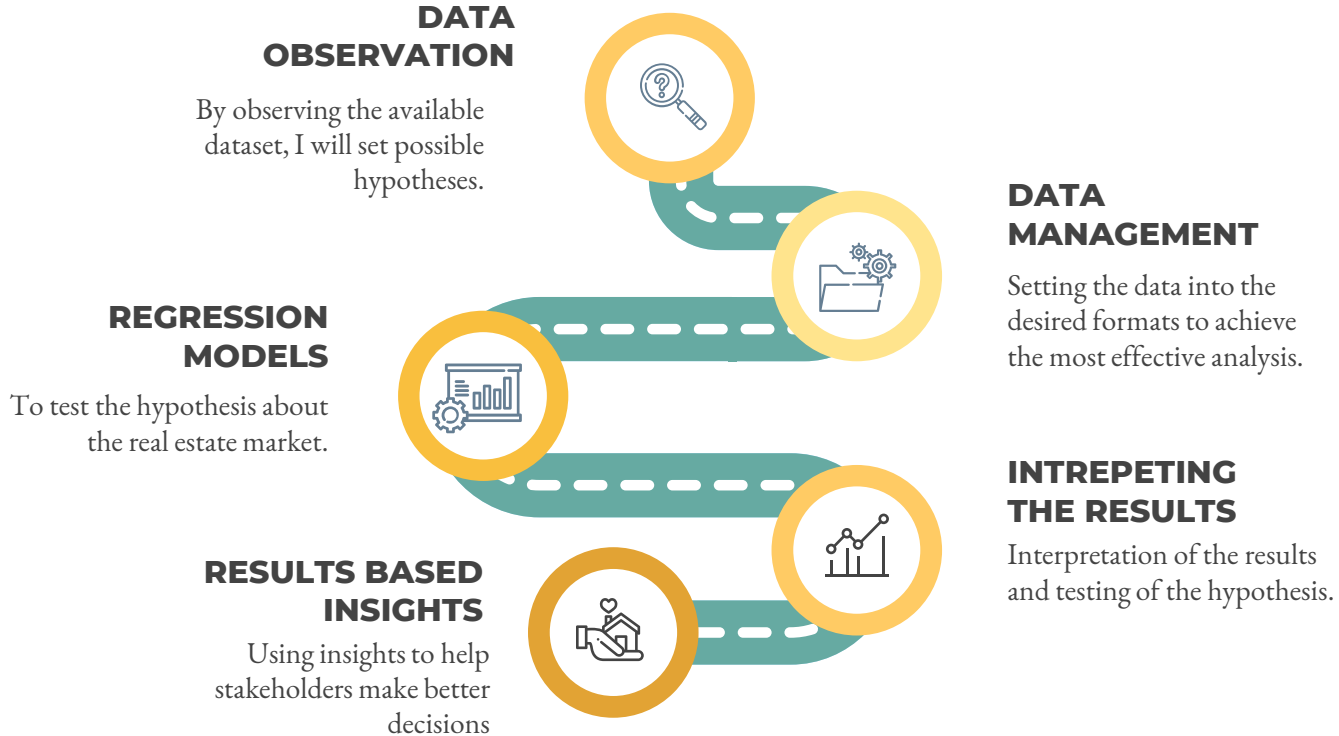


What was the requirement for the homework/research at hand?

- **Real data challenge: specify and estimate a model for price prediction of apartments in Sofia**
- **Assignment output: a brief research report (ppt presentation) with your major findings.**



STAGES OF THE ANALYSIS



From the available dataset, there were 14 variables available; 12 were used (some with data management, some without). Fewer of the baseline ones were used to test the hypothesis and conduct the regression models.

Variables: (Name – operation – description) * - modified, X - excluded, T - transformation, D - dummy, N - none

Type - *, D - type of property

Regiid - X - region of the property

Townid - X - town of the property

Quartid - *, T - the neighborhood of the property

Price (Euro) - T (ln) - the advertised price for the property

Euro/sq.m. - t (ln) - the price per sq. meter

Area - T (mean, med per neig.) - area/size of the property

Constr. - *, D - type of construction of the property

Year - t (mean med per neig.) - construction year

Floor - N - floor of the property

TEC - * - central heating variable (ДА/HE) - transformed to 1/0

Published or Renewed & Valid date - * - a made a operational variable from both of them named Listing days.

Seller - D - Either Agency or Individual (personal_seller)

P.S If there was data if the apartment had furniture and was remoted would have made the whole research more precise.



Data Analysis

Data Management

The following changes were made

Type – was modified to have 5 categories instead of 7 unique categories. With 2 extra dummies

Quartid – evolved into 4 different variables

NS coordinate

WE coordinate

Distance from center

Metro

Price (Euro) & Euro/sq.m. – were transformed into natural logs for regression purposes.

Area & Year – had Mean & Median calculated per neighborhood (to assess premiums.)

Constr. – construction variable was split into three dummies for each category (Panel, NPK, Brick).

Heating – was modified to be a logical instead of char. type.

Personal Seller - dummy was used to see does agencies affect price



Quartid dummies - calculation

A prop map of Sofia's neighborhoods was done in Excel.
Each square is 500m x 500m

Using the map, each neighborhood's central point was calculated and set to be the NS = [-9,9] and WE = [-9,9] coordinate.

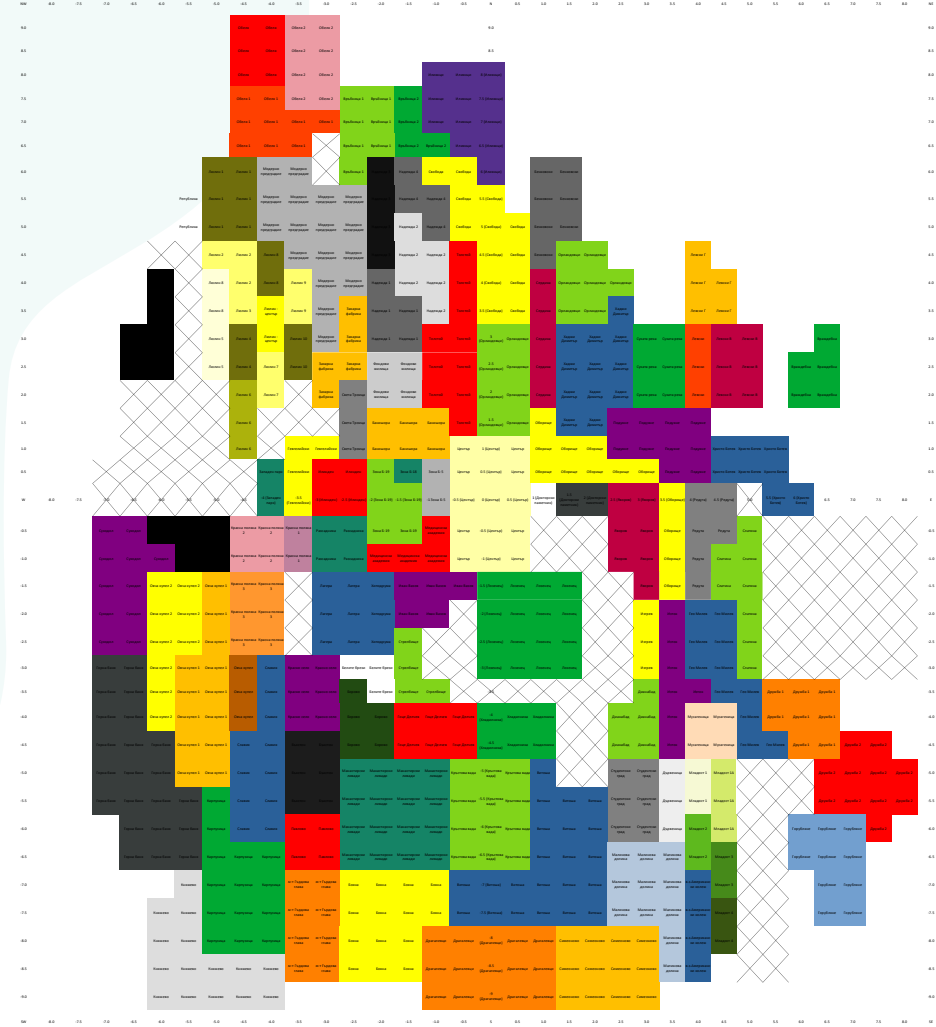
Using the Pythagorean theorem, the aerial distance from the center was calculated.

Distance from Center = $\sqrt{NS^2 + WE^2}$

When compiling the map each neighborhood was assigned if it had metro access or not.

NS formula for Geo Milev
=-(ROUND((ROW(\$MAP.AA25)
+ ROW(\$MAP.AD30)) / 2, 1) - 21) / 2

WE formula for Geo Milev
=(ROUND((COLUMN(\$MAP.AA25) +
COLUMN(\$MAP.AD30)) / 2, 1) - 19) / 2



Regression Analysis

DIAGNOSTIC model

Using $\ln(\text{euro_sq_meter})$ to determine the right regressors.



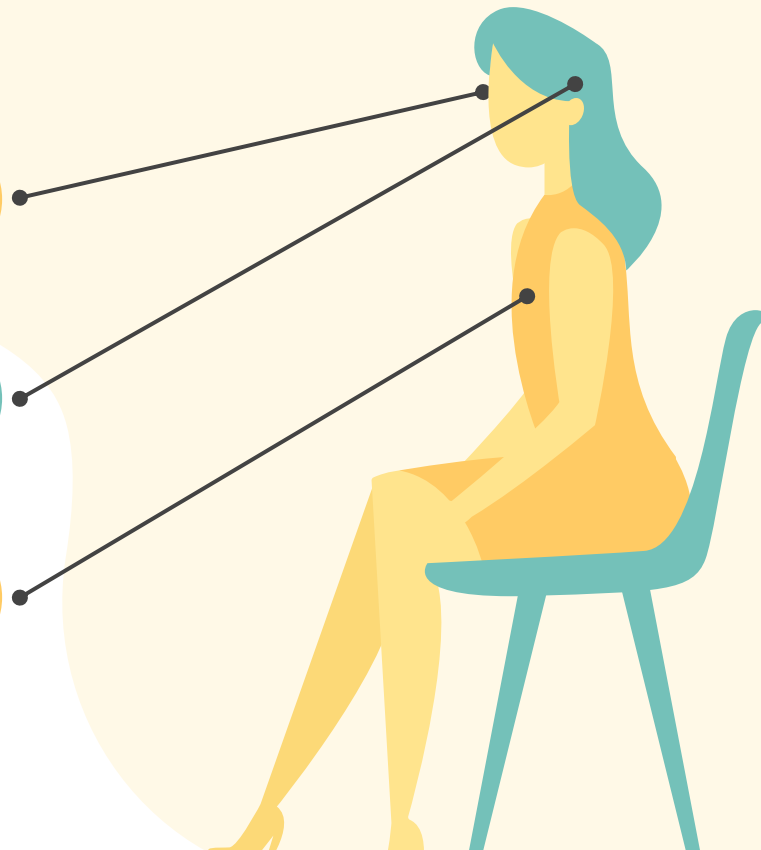
BIG DATA model

Predicting $\ln(\text{price})$ using all economically sound and econometrically appropriate regressors.



NO BIG DATA model

Predicting $\ln(\text{price})$ using layman-accessible regressors.



Diagnostic Model

Test Model

Sample restrictions

The sample was restricted with construction type brick and access to metro in order to diagnostisize the more crutial non dummie variables.

Assesment of proper independed variables and testing the hypotesis that location, proximity to the center & access to public transport are the crutial factors influencing the price.

Neighborhood BIAS was present

Half of the Neighborhoods had approximately 85% of all observations in the data.



Diagnostic Model

Equation: BRICK_AND_METRO_LN Workfile: 7.4-SOFIA PROPERTIES::Untitled\

View Proc Object Print Name Freeze Estimate Forecast Stats Resids

Dependent Variable: LN_EURO_SQ_M

Method: Least Squares

Date: 11/04/25 Time: 19:09

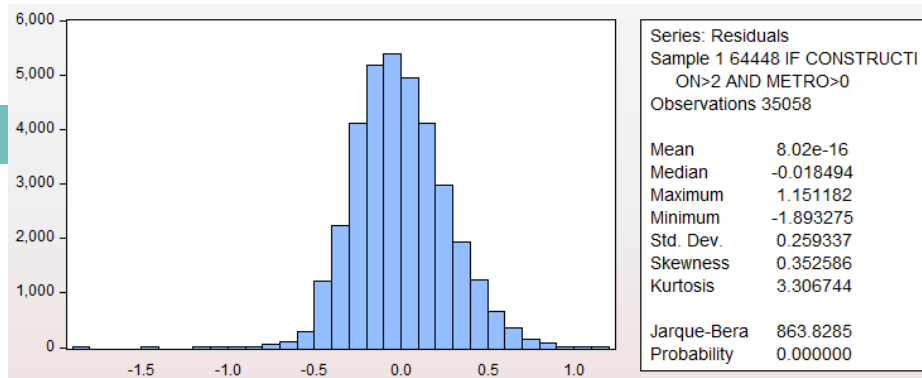
Sample: 1 64448 IF CONSTRUCTION>2 AND METRO>0

Included observations: 35058

Huber-White-Hinkley (HC1) heteroskedasticity consistent standard errors and covariance

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	6.892068	0.002450	2812.835	0.0000
AREA_MINUS_AVG_Q	-0.002169	4.65E-05	-46.62015	0.0000
WE	0.032720	0.000513	63.79485	0.0000
HEATING	0.143238	0.002858	50.12006	0.0000
OBSV_WEIGHT	2.459667	0.043441	56.62128	0.0000
BIG_PROPERTY	0.095028	0.004814	19.73929	0.0000

R-squared	0.282174	Mean dependent var	7.037817
Adjusted R-squared	0.282071	S.D. dependent var	0.306095
S.E. of regression	0.259356	Akaike info criterion	0.138941
Sum squared resid	2357.791	Schwarz criterion	0.140389
Log likelihood	-2429.490	Hannan-Quinn criter.	0.139402
F-statistic	2755.753	Durbin-Watson stat	1.377366
Prob(F-statistic)	0.000000	Wald F-statistic	2801.220
Prob(Wald F-statistic)	0.000000		



Variance Inflation Factors

Date: 11/06/25 Time: 12:36

Sample: 1 64448 IF CONSTRUCTION>2 AND METRO>0

Included observations: 35058

Variable	Coefficient Variance	Uncentered VIF	Centered VIF
C	6.00E-06	3.345571	NA
AREA_MINUS_AVG_Q	2.16E-09	1.551136	1.508193
WE	2.63E-07	1.195369	1.059203
HEATING	8.17E-06	1.622346	1.002111
OBSV_WEIGHT	0.001887	2.310113	1.056384
BIG_PROPERTY	2.32E-05	1.762064	1.512420

Big Data Model

Intended use by RE Agencies and Serious Brokers

Most RE Agencies have sample data about the postings in each area. There for the use of the Means and Medians of each neighbourhood was used, to caculate the diviations of the area and year of construction to set most efficient premiums.

Main Model

Maximizing the prediction power and overall stability model of price setting by the regressors available.

Full Dummy utalization

Dummy utilization was essential for specifying various nuances in properties.



Big Data Model

Equation: PRICE_EURO_LN Workfile: 7.4-SOFIA PROPERTIES::Untitled\

View Proc Object Print Name Freeze Estimate Forecast Stats Resids

Dependent Variable: LN_PRICE_EURO_

Method: Least Squares

Date: 11/05/25 Time: 13:52

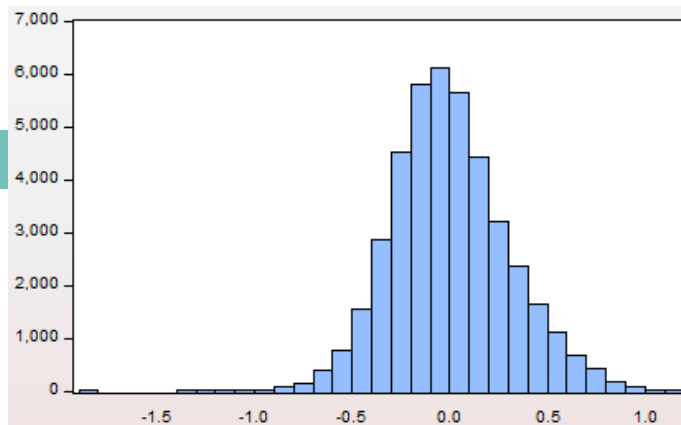
Sample (adjusted): 1 64447

Included observations: 41979 after adjustments

Huber-White-Hinkley (HC1) heteroskedasticity consistent standard errors and covariance

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	11.38022	0.003136	3628.781	0.0000
AREA_MINUS_AVG_Q	0.006809	6.13E-05	111.1161	0.0000
YEAR_MINUS_AVG_Q	-0.001517	9.84E-05	-15.41974	0.0000
NS	-0.026457	0.000362	-73.00101	0.0000
WE	0.007455	0.000510	14.61267	0.0000
HEATING	0.079764	0.003194	24.97323	0.0000
BIG_PROPERTY	0.090847	0.005381	16.88273	0.0000
STUDIO	-0.440842	0.005564	-79.23574	0.0000
METRO	0.033027	0.003025	10.91830	0.0000
PERSONAL_SELLER	0.062449	0.013608	4.589275	0.0000

R-squared	0.549397	Mean dependent var	11.47990
Adjusted R-squared	0.549300	S.D. dependent var	0.437731
S.E. of regression	0.293867	Akaike info criterion	0.388858
Sum squared resid	3624.347	Schwarz criterion	0.390917
Log likelihood	-8151.937	Hannan-Quinn criter.	0.389508
F-statistic	5685.627	Durbin-Watson stat	1.575549
Prob(F-statistic)	0.000000	Wald F-statistic	5393.225
Prob(Wald F-statistic)	0.000000		



Series: Residuals
Sample 1 64447
Observations 41979

Mean -1.88e-15
Median -0.021067
Maximum 1.141787
Minimum -1.870675
Std. Dev. 0.293835
Skewness 0.270725
Kurtosis 3.295634

Jarque-Bera 665.6593
Probability 0.000000

Variance Inflation Factors

Date: 11/06/25 Time: 12:40

Sample: 1 64448

Included observations: 41979

Variable	Coefficient Variance	Uncentered VIF	Centered VIF
C	9.84E-06	5.186017	NA
AREA_MINUS_AVG_Q	3.76E-09	1.797493	1.788434
YEAR_MINUS_AVG_Q	9.68E-09	1.083162	1.077438
NS	1.31E-07	1.814505	1.303426
WE	2.60E-07	1.231926	1.156494
HEATING	1.02E-05	1.975999	1.194468
BIG_PROPERTY	2.90E-05	1.741081	1.520259
STUDIO	3.10E-05	1.314906	1.204729
METRO	9.15E-06	2.865463	1.125032
PERSONAL_SELLER	0.000185	1.012984	1.000280

NO Big Data Model

Simplified version

Percentage change in price because of the simple regressors used.

Practical Model

Layman guide for getting a guide for pricing properties.

Most crucial Dummy Variables

Property access to central heating, metro station and dealing with personal sellers.



NO Big Data Model

Equation: NO_BIG_DATA Workfile: 7.4-SOFIA PROPERTIES::Untitled\

View Proc Object Print Name Freeze Estimate Forecast Stats Resids

Dependent Variable: LN_PRICE_EURO_

Method: Least Squares

Date: 11/05/25 Time: 13:56

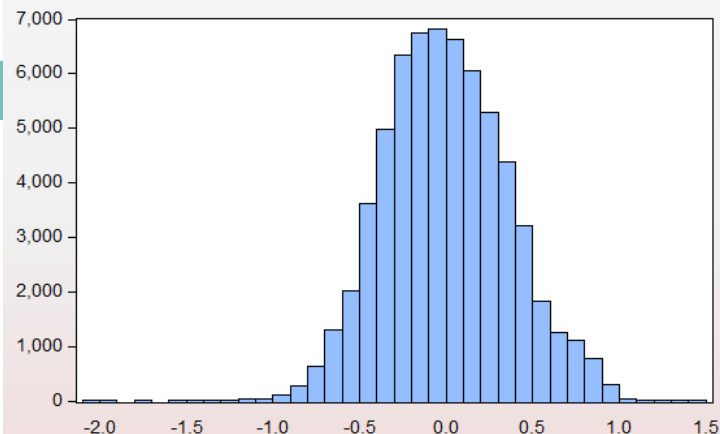
Sample (adjusted): 1 64447

Included observations: 63822 after adjustments

Huber-White-Hinkley (HC1) heteroskedasticity consistent standard errors and covariance

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	11.35934	0.003222	3525.951	0.0000
NS	-0.023751	0.000379	-62.67137	0.0000
WE	0.010264	0.000507	20.23546	0.0000
HEATING	0.086208	0.003052	28.24663	0.0000
BIG_PROPERTY	0.472290	0.003847	122.7770	0.0000
STUDIO	-0.690954	0.004682	-147.5770	0.0000
METRO	0.035658	0.003071	11.61088	0.0000
PERSONAL_SELLER	0.051736	0.013759	3.760197	0.0002

R-squared	0.344602	Mean dependent var	11.48783
Adjusted R-squared	0.344530	S.D. dependent var	0.439176
S.E. of regression	0.355562	Akaike info criterion	0.769889
Sum squared resid	8067.622	Schwarz criterion	0.771025
Log likelihood	-24559.92	Hannan-Quinn criter.	0.770240
F-statistic	4793.251	Durbin-Watson stat	1.249656
Prob(F-statistic)	0.000000	Wald F-statistic	7362.901
Prob(Wald F-statistic)	0.000000		



Series: Residuals
Sample 1 64447
Observations 63822

Mean 1.79e-15
Median -0.016316
Maximum 1.415819
Minimum -2.095631
Std. Dev. 0.355542
Skewness 0.163531
Kurtosis 2.890129

Jarque-Bera 316.5591
Probability 0.000000

Variance Inflation Factors

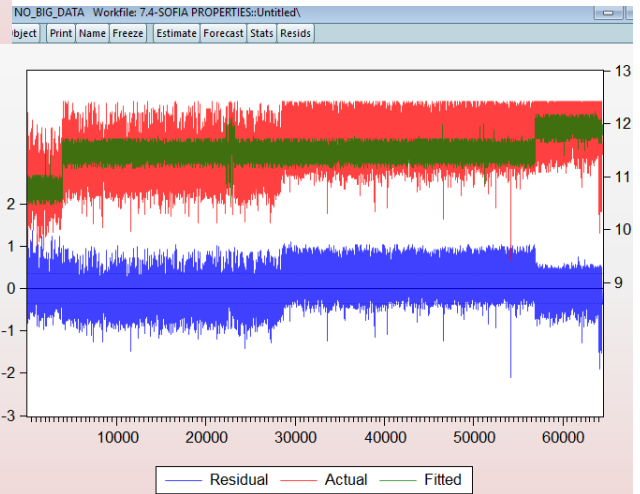
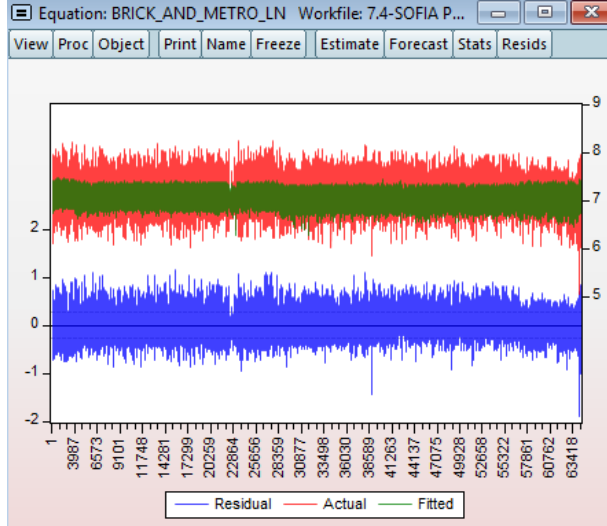
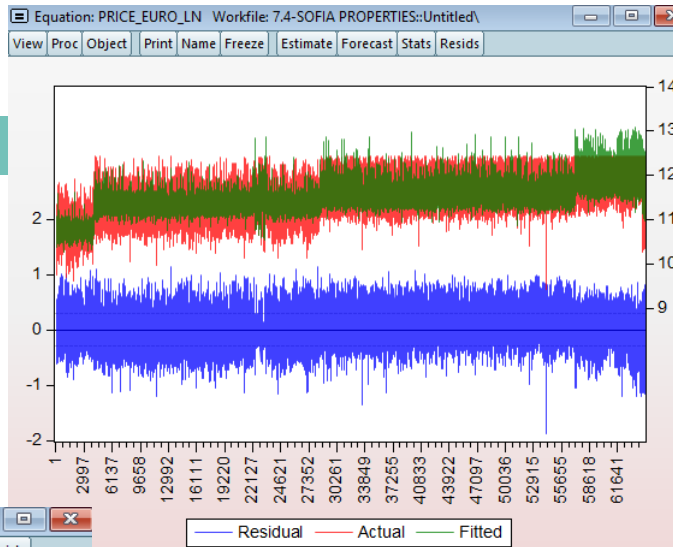
Date: 11/06/25 Time: 12:42

Sample: 1 64448

Included observations: 63822

Variable	Coefficient Variance	Uncentered VIF	Centered VIF
C	1.04E-05	5.529184	NA
NS	1.44E-07	1.785686	1.337066
WE	2.57E-07	1.222231	1.172867
HEATING	9.31E-06	2.055283	1.151534
BIG_PROPERTY	1.48E-05	1.230844	1.040329
STUDIO	2.19E-05	1.143439	1.029053
METRO	9.43E-06	3.180470	1.123419
PERSONAL_SELLER	0.000189	1.011646	1.000371

Residual Analysis



Quick Summary



Location

Location of the properties is influenced aproximetly by
-2% for every 1km - North off the Center
+1% for every 1km – East off the Center



Metro

Properties that have a Metro station nearby
results in an increase of price by around 3.5%.



Cental Heating & Personal Seller

Central Heating increases the price by around 8%
& Personal Seller increases the price by around 6%

Thank you for your attention

Q&A

“Real estate is all about location, location, location.”

— **Baron** Harold Samuel , founder of
Land Securities (UK)

